

Unit 12, Lesson 1: Creating Proportions



Benchmark

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.



Learning Target

- ☐ I can create proportions to represent a situation.



12.1.1 Warm-Up: Math Flop

The advertisement encourages making DiGiorno pizza a part of your weekly meals.

1. What do you notice about the advertisement?





12.1.2 Exploration: Proportional Thinking

A photograph is enlarged to make a poster. Measurements are given in centimeters (cm).

16 cm



10 cm

?



25 cm

Abigail used her pencil and measured the photo's width using her finger to mark off 10 cm on the pencil. She then used the pencil to estimate the height using the mark on her pencil.

1. Discuss with your partner why Abigail's strategy could lead to an incorrect answer.

Mariana and Yousef used different strategies to find the height of the poster. Review their solution strategies. Then, work with your partner to answer the questions that follow.

Mariana's Work	Yousef's Work
$\frac{25 \text{ cm}}{10 \text{ cm}} \div \frac{10}{10} = \frac{2.5}{1} = 2.5$ Scale factor = 2.5 $16 \text{ cm} \times 2.5 = 40 \text{ cm}$ The height of the poster is 40 cm.	$\frac{\text{height}}{\text{width}} \quad \frac{16 \text{ cm}}{10 \text{ cm}} = \frac{?}{25 \text{ cm}}$ $\frac{16 \text{ cm}}{10 \text{ cm}} \times \frac{2.5}{2.5} = \frac{40 \text{ cm}}{25 \text{ cm}}$ The height of the poster is 40 cm.

2. How are Mariana and Yousef's solution strategies similar?
3. How are their solution strategies different?



12.1.3 Bring It Together: Proportional Relationships



In a **proportional relationship**, the values for one quantity are each multiplied by the same number to get the values for the other quantity.

A **proportion** sets two or more equivalent ratios in such relationships as equal to each other and can be used to solve problems.



A **unit rate** is a ratio comparing a number of units of one quantity to 1 unit of a second quantity.

- Use the words and phrases in the bank below to complete the description of Mariana and Yousef's work.

is equal to proportion ratio unit rate

The photograph and the poster are in a **proportional relationship** because the poster is a scaled version of the photograph. Mariana found the scale factor, which is a _____, and multiplied it by the height of the photo to find the height of the poster. Yousef set up a _____ stating the _____ of the height to width of the photograph _____ the ratio of the height to width of the poster.



12.1.4 Guided Instruction: Proportional Thinking

Proportions can be set up in different ways using the relationships between each quantity.

Four different proportions that can be used to find the height of the poster are shown.

A photo to poster $\frac{\text{height}}{\text{width}} \left \frac{16 \text{ cm}}{10 \text{ cm}} = \frac{?}{25 \text{ cm}} \right.$ $\frac{16 \text{ cm}}{10 \text{ cm}} \times \frac{2.5}{2.5} = \frac{40 \text{ cm}}{25 \text{ cm}}$	B poster to photo $\frac{\text{width}}{\text{height}} \left \frac{25 \text{ cm}}{?} = \frac{10 \text{ cm}}{16 \text{ cm}} \right.$ $\frac{25 \text{ cm}}{40 \text{ cm}} = \frac{10 \text{ cm}}{16 \text{ cm}} \times \frac{2.5}{2.5}$
C width to height $\frac{\text{photo}}{\text{poster}} \left \frac{10 \text{ cm}}{25 \text{ cm}} = \frac{16 \text{ cm}}{?} \right.$ $\frac{10 \text{ cm}}{25 \text{ cm}} \times \frac{1.6}{1.6} = \frac{16 \text{ cm}}{40 \text{ cm}}$	D height to width $\frac{\text{photo}}{\text{poster}} \left \frac{16 \text{ cm}}{?} = \frac{10 \text{ cm}}{25 \text{ cm}} \right.$ $\frac{16 \text{ cm}}{40 \text{ cm}} = \frac{10 \text{ cm}}{25 \text{ cm}} \times \frac{1.6}{1.6}$

All of the proportions show that the ratios of the dimensions of the photograph and the dimensions of the poster are equivalent.

Complete the statements based on the proportions.

- The dimensions of the poster are ☐ 1.6 ☐ 2.5 times larger than the photograph.
- The height of both the photograph and poster is ☐ 1.6 ☐ 2.5 times larger than the width.



12.1.5 Guided Practice: Proportional Thinking

The city of Tampa collected sample data on pet ownership to use when determining whether or not it needed to build additional dog parks. A city parks officer randomly sampled 200 households in Tampa and found that 126 of the households had a dog. Using the results of the sample, determine how many households have a dog if there are 153,918 households in Tampa.

- Complete the statements to describe the relationships that are being compared.
 - The number of households with a dog is being compared to the _____.
 - The data sample is being compared to the _____ of Tampa.
- Two proportions that represent this situation are below, but one of them has an error.

Proportion A	Proportion B
$\frac{126}{200} = \frac{153,918}{x}$	$\frac{153,918}{200} = \frac{x}{126}$

Proportion ☐ A ☐ B is not set up correctly. The proportion should be _____ = _____.

- Find the number of households expected to have a dog.



12.1.6 Collaboration: Creating Proportions

- Javad drove his car to Canada. He was driving on a road and saw the sign shown to the right. According to his speedometer, Javad was driving 75 miles per hour when he was pulled over for speeding. Canadian speed limits are given in kilometers (km) per hour because they use the metric system. There are approximately 1.61 kilometers in 1 mile.



- Create two different proportions that could be used to find Javad's speed in kilometers per hour.

Proportion A	Proportion B

- How fast was Javad traveling in kilometers per hour?

2. An architect made a scale drawing of a house as shown.



The scale for the drawing is $\frac{1}{4}$ inch = 6 feet.

- a. Create two different proportions that could be used to find the actual length of the house represented by the drawing.

Proportion A	Proportion B

- b. What is the length of the house represented by the drawing?



12.1.7 Wrap-Up: Error Analysis

Jacksonville and Gainesville are approximately 1.4 inches (in.) apart on a scaled map of Florida. The scale used for the map is 2 in. = 90 miles (mi).

Three students set up proportions to find the actual distance between Jacksonville and Gainesville, but two of them made errors.

1. For each proportion, check whether it is correct or incorrect. If it is incorrect, fix the error by changing one of the ratios in the proportion.

Lily's Proportion	Eli's Proportion	Mia's Proportion
$\frac{2}{x} = \frac{90}{1.4}$	$\frac{1.4}{2} = \frac{90}{x}$	$\frac{1.4}{x} = \frac{2}{90}$
☐ correct ☐ incorrect	☐ correct ☐ incorrect	☐ correct ☐ incorrect